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Product Specification

Product: InnoCAM_DCM_IMX586PDAF

Part Number: INV-IMX586AF-48MP

Revision: Rev 1.2

Document No: INV421

REVISION HISTORY

Revision	Description of change	Changed by	Date
1.0	Initial Draft	Jamie Lynn	03/21/2022
1.1			
1.2			
1.3			

APPROVAL

Company	Name	Signature	Date
InnoWave Design LLC	Tony Reed		07/09/2022
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1. General

The INV-IMX586AF-48M is an auto focus camera module with a diagonal 8.000 mm (Type 1/2.0) Sony IMX586 48 Mega-pixel CMOS active pixel type stacked image sensor with a square pixel array. It adopts Sony's back-illuminated and stacked CMOS image sensor to achieve high speed image capturing by column parallel A/D converter circuits and high sensitivity and low noise image (comparing with conventional CMOS image sensor) through the backside illuminated imaging pixel structure. R, G, and B pigment primary color mosaic filter is employed. It operates with five power supply voltages: analog 2.9 V and 1.8V, digital 1.1 V, PLL-PHY 1.1V and 1.8 V for input/output interface and achieves low power consumption. The camera module has an auto focus lens, holder, VCM and FPC.

1.1. Specifications

Sensor Make and Model	IMX586-AAJH5-C
Sensor Type	CMOS
Resolution	48MP
Active array size	5360x4032
Pixel Size	0.80um x 0.80um
Module Size	10.8x10.3x6.5mm
Optical size	1/2.0"
Output Format	10-bit RAW RGB data
Chroma	Color
Image Size	Diagonal 8 mm (Type 1/2.0)
Total Number of pixels	8032 (H) × 6248 (V) approx. 50.18 M pixels
Number of effective pixels	8032 (H) × 6088 (V) approx. 48.89 M pixels
Number of active pixels	8000 (H) × 6000 (V) approx. 48.00 M pixels
Chip Size	7.504 mm (H) × 5.659 mm (V)
Substrate material	Silicon
Frame Rate	30fps@ Full resolution 1000fps@4000x3000 120fps@2x2 Adjacent Pixel Binning (16:9) 240fps@2x2 Adjacent Pixel Binning 2000x11282(16:9) 480fps@2x2 Adjacent Pixel Binning 720 crop (16:9)
Sensor CRA	35 degrees
Guaranteed Temperature Ranges	Operating -20C to +70C Storage -30 to +80C Performance -20C to +60C
Lens Manufacturer	
Lens Model	
Construction	6P
Lens Type	Auto focus
Field of View (FOV) degrees	Diagonal 85
Aperture (F#)	1.79

INV-IMX586AF-48M

EFL	4.74mm
Distortion (TV)	<1.5%
Focus Range	8cm - inf

Table 1: Specifications

1.2. Sensor Features

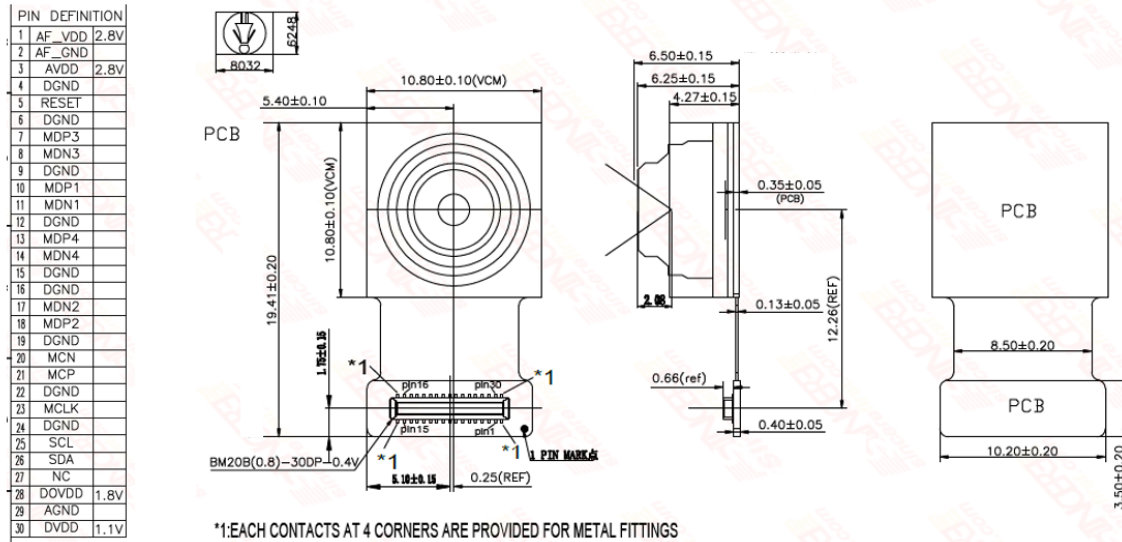
Back-illuminated and stacked CMOS image sensor
Quad Bayer Coding color filter arrangement
Phase Detection Auto Focus (PDAF)
High Frame Rate 30fps@Full resolution (QBC Re-mosaic) / 30fps@QBC-HDR / 120fps@2x2 Adjacent Pixel Binning (16:9) / 240fps@2x2 Adjacent Pixel Binning V2H2(16:9)
High signal to noise ratio(SNR)
Dual sensor synchronization operation
Built-in 2D Dynamic Defect Pixel Correction
Lens Shading Correction (LSC)
Built-in temperature sensor
Output video format of RAW10/8, COMP8
QBC Re-mosaic function
QBC HDR function
Two PLLs for independent clock generation for pixel control and data output interface
CSI-2 serial data output
MIPI D-PHY 2lane/4lane, Max. 2.5Gbps/lane, D-PHY spec. ver. 1.2 compliant
MIPI C-PHY 1/2/3trio, Max 2.5Gbps/Trio, C-PHY spec ver. 1.0 compliant
2-wire serial communication (Supports I2C "Fast mode" and "Fast-mode Plus")
28K bit of OTP ROM for users

Table 2: Sensor Features

1.3. Applications

- Cellular Phones
- Tablet PCs

1.4. Layout



Revision Record							InnoCAM Beyond Expectations By InnoWave		
			Sensor Size	1/2"	Sensor	IMX586-AAIH5-C			
			F. NO	1.79	Array Size	8032x6248	Scale 1:1	Project Name:	DCM_IMX586AF_48MP
			DFOV	85° +/- 3	Distortion	<1.5%	Unit : mm	Drawing No.	
AO		Initial Drawing	Focus Range	8cm-inf	Lens Construction	6P	Ver	DWN By	CHK By: App By:
VER.	Date	Description	EFL	4.74mm			Page 1/1	Jamie Lyn	Tony Reed Tony Reed

Figure 1: camera module assembly layout

1.5. Lens

Figure 2: Lens assembly

2. Electrical

2.1. Absolute Maximum Ratings

Item	Symbol	Ratings	Unit	notes
Supply voltage (analog1)	VANA1	-0.3 to +4.2	V	refer to VSS level
Supply voltage (analog2)	VANA2	-0.3 to +2.52	V	
Supply voltage (digital)	VDIG	-0.3 to +1.54	V	
Supply voltage (interface)	VIF	-0.3 to +2.52	V	
Input voltage (digital)	VI	-0.3 to +2.52	V	
Output voltage (digital)	VO	-0.3 to +2.52	V	

Table 3: Absolute Maximum Ratings

2.2. Recommended Operation Voltage

Item	Symbol	Ratings	Unit	notes
Supply voltage (analog1)	VANA1	2.9 ± 0.1	V	refer to VSS level
Supply voltage (analog2)	VANA2	1.8 ± 0.1	V	
Supply voltage (digital)	VDIG	1.1 ± 0.1	V	
Supply voltage (interface)	VIF	1.8 ± 0.1	V	

Table 4: Recommended operation voltage

3. Environment Requirements

3.1. Operating Temperature

The camera module shall be fully functional when ambient temperature is between -20°C to 60°C with image quality remaining stable. Test duration is 24 hours.

3.2. Storage Temperature

The camera module shall withstand storage temperatures between -30°C to 70°C. Test duration is 48 hours.

3.3. Humidity

The camera module shall withstand humidity at or below 90% RH under non-condensing conditions for 24 hours.

3.4. Thermal Shock

The camera module shall withstand the following temperatures (with humidity off)

-30°C to 70°C

20 min cycles (10 min dwell, 5 min ramp, 10 min dwell)

3.5. High Temperature Test

60°C, humidity off, 24 hours

3.6. Low Temperature Test

-20C, humidly off, 24 hours

Stable image is -30°C to 70°C junction temperature. The sensor functions but image quality may be noticeably different at temperatures outside of stable image range. Image quality remains stable between 0°C to 50°C.

4. Reliability Requirements

4.1. Drop Test

The camera module shall withstand a 1.2m Drop in packaging onto Concrete (12 drops) Random Positions

4.2. Random Vibration

The camera module shall withstand vibration of the following conditions

Frequency range: 50Hz

Amplitude: 2mm Duration 10 minutes for each position

Test all 3 axes (X, Y, Z)

4.3. Salt Fog Test

Condition: 5%nacl solvent Test duration: 24H

4.4. ESD (Electronic Discharge)

The camera module shall withstand Electrostatic Discharge of

8KV Contact Discharge

12KV Air Discharge

10 Times for a Second

5. Product Performance Verification

To verify the camera module performance, the following tests will be conducted at either the factory during production or as an initial qualification characterization in either the factory laboratory or at the InnoWave laboratory.

5.1. Electrical Parameters

Parameter	Test Frequency
Current consumption – Standby	Initial Qualification
Current consumption – Idle	Initial Qualification
Current consumption – Viewfinder	Initial Qualification
Current consumption – Capture	Initial Qualification

Table 5: Electrical parameter measurements

5.2. Mechanical Parameters

Parameter	Test Frequency
X Dimension (mm)	Initial Qualification
Y dimension (mm)	Initial Qualification
Z Dimension (mm)	Initial Qualification

Table 6: Mechanical parameter measurements

5.3. Environmental and Reliability Test Parameters

Parameter	Test Frequency
Thermal Shock	Initial Qualification
Humidity	Initial Qualification
High Temperature Test	Initial Qualification
Low Temperature Test	Initial Qualification
Drop Test	Initial Qualification
Random Vibration Test	Initial Qualification
Salt Fog Test	Initial Qualification
ESD Test	Initial Qualification

Table 7: Environmental and Reliability parameter measurements

6. Product Identification TBD

All modules will be marked with an identification number using laser marking or bar code label.

7 Packaging

The package will prevent damage to the components during transport and will be suitable for electrostatic-sensitive devices. The single camera modules shall be delivered in a reusable tray of anti-static plastic material. Several cameras shall be packed in one tray.

The tray has separate holders for each camera module.

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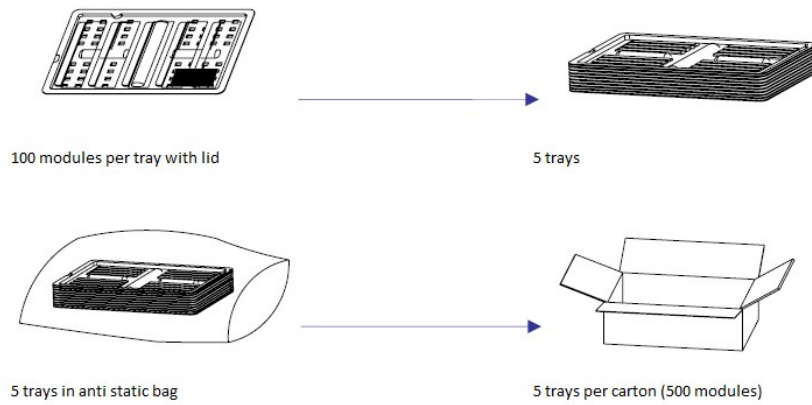


Figure 4: Packaging Example